

Quick_Activity_1.pdf

A notebook costs ₱45, and a pen costs ₱12. A student buys several notebooks and pens and pays a certain amount. Develop an algorithm that calculates the total cost and change.

Problem Analysis

Inputs	Processes	Outputs
Quantity of notebook, quantity of pens, payment amount	Calculate notebook cost, calculate pen cost, calculate total cost, calculate change	Total cost Change

Decomposition

1. Input the quantity of notebooks and pens
2. Input the payment
3. Calculate the notebook cost
4. Calculate the pen cost
5. Calculate the total cost
6. Calculate the change
7. Display the total cost
8. Display the change

Pseudocode

START

 READ notebookPrice, penPrice, paymentAmount

 notebookCost $\leftarrow$ quantity $\times$ 45

 penCost $\leftarrow$ quantity $\times$ 12

 totalCost $\leftarrow$ notebookCost + penCost

 change $\leftarrow$ payment – totalCost

 DISPLAY totalCost, change

END

Dry-Run Calculation

Test Case 1

Input:

- Notebook: 3
- Pens: 2
- Payment: ₱200

Step	Variable	Calculation	Result
1	notebookCost	$3 \times \text{₱}45$	₱135
2	penCost	$2 \times \text{₱}12$	₱24
3	totalCost	$\text{₱}135 + \text{₱}24$	₱159
4	change	$\text{₱}200 - \text{₱}159$	₱41

Test Case 2

Input:

- Notebook: 2
- Pens: 3
- Payment: ₱150

Step	Variable	Calculation	Result
1	notebookCost	$2 \times \text{₱}45$	₱90
2	penCost	$3 \times \text{₱}12$	₱36
3	totalCost	$\text{₱}90 + \text{₱}36$	₱126
4	change	$\text{₱}150 - \text{₱}126$	₱24